

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID: SIH26094

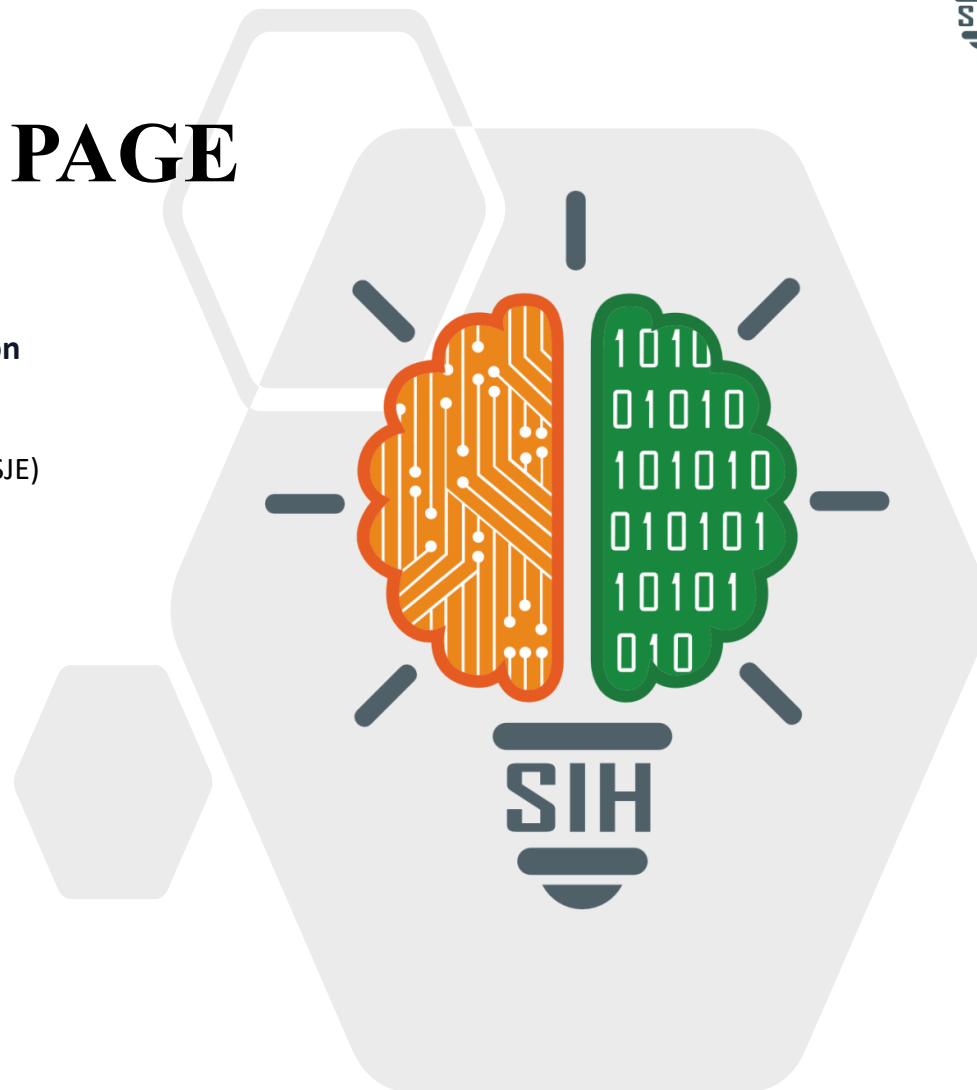
Title: AI-Powered Dynamic Mental Health Monitoring and Distress Prediction System for Victims of Atrocities

Theme / Technology Bucket: MedTech / BioTech / HealthTech (Software)

PS Category: Software | Ministry: Ministry of Social Justice and Empowerment (MoSJE)

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IDEA TITLE



- Project Identity & Greek Origin: KEFI AI (Kinematic Emotion & Forecasting Intervention System) — derived from Kéfi (Κέφι): spirit of joy, emotional vitality, resilience & inner healing after trauma.
- Continuous Post-Complaint Monitoring: Proactively monitors crime victims (SC/ST PoA Act 1989 / NHAA 14566) across investigation, court trial, legal delay, compensation, and rehabilitation phases rather than waiting for crisis requests.
- Multichannel Interaction Hub: Periodic check-ins via IVRS 14566 calls, Chatbot, SMS, Mobile App, and Web Portal with digital OTP consent & AES-256 encryption.
- Dual-Risk Assessment Engine: 1) Immediate Physical Threat Check: Screens for active intimidation language. 2) Dynamic Distress Score (DDS): Multi-signal mathematical metric ($DDS = w1 \cdot E + w2 \cdot V + w3 \cdot B + w4 \cdot H + w5 \cdot C + w6 \cdot T$).
- 3-Tier Risk Bands & Action Matrix: ● Green/Yellow (0–30): Routine wellness check-ins & self-help tools. ● Orange (31–60): Focused check-ins & scheduled tele-counselling. ○ Red (61–100): Real-time alert to District Magistrate (DM) / SP dashboard, witness protection & crisis team dispatch.

- Multimodal Data Analysis Engine: NLP Sentiment & Emotion AI (Transformer BERT for fear valence & threat keywords 'dhamki', 'withdraw', 'court'), Vocal Acoustic Prosody (pitch tremor Hz, jitter %, pause duration), and Behavioral Check-in Gaps.
- Mathematical Dynamic Distress Scoring (DDS): $DDS = 0.25 * E + 0.20 * V + 0.15 * B + 0.15 * H + 0.10 * C + 0.15 * T$ (Emotion, Voice Stress, Behavior, Historical Trend, Case Factors, Threat Language).
- Longitudinal Trend & 72h Forecasting: Calculates distress velocity (dDDS/dt) over past 7 days to predict psychological crisis 72 hours before escalation occurs.
- Explainable AI (SHAP / LIME XAI): Provides exact point attributions (+21 threat, +18 fear, +14 velocity, +12 missed check-ins, +9 voice stress) for transparent, audit-ready officer decisions.
- Technology Stack & Architecture: Python FastAPI backend, PyTorch, React + Vite frontend, SQLite/PostgreSQL, Web Speech API, DPDP 2023 Security.

FEASIBILITY AND VIABILITY



- Operational Feasibility: Integrates seamlessly with existing MoSJE National Helpline Against Atrocities (NHAA 14566), welfare officers, and police liaisons without overloading staff.
- Low-Bandwidth & Offline Adaptability: Low Internet Mode allows offline grounding audio, low-data layout, and basic SMS fallback for remote rural atrocity victims.
- Risk 1: Alarm Fatigue & False Positives: Mitigation: Dual-layer verification combining AI Distress Score with Field Welfare Officer ground observation notes.
- Risk 2: Data Privacy & Security Concerns: Mitigation: Fully DPDP Act 2023 compliant with end-to-end AES-256 encryption, victim data consent management, and auto-purge options.
- Risk 3: Multilingual & Literacy Barriers: Mitigation: Multilingual conversational AI supporting Tanglish, Tamil, Hindi, and English + Silent Mode body language & posture detection.

IMPACT AND BENEFITS



- Target Beneficiaries (SC/ST PoA Act 1989): Victims of atrocity (rape/gang rape, grievous hurt, murder, arson), witnesses facing trial intimidation, and affected families under SC/ST (Prevention of Atrocities) Act 1989.
- Early Mental Crisis Prevention: Identifies escalating psychological distress before severe trauma, relapse, or self-harm occurs, providing timely crisis tele-counselling.
- Victim Empowerment & Self-Help: Provides 5-4-3-2-1 grounding, box breathing, legal rights pamphlets, and transparent feedback loops.
- Strengthened Justice System Trust: Proactive state care rebuilds victim confidence, reducing witness retraction and trial dropouts.
- Evidence-Based Governance: Real-time district, state, and national MoSJE dashboards for District Magistrates (DM) and Superintendents of Police (SP) to prioritize high-risk cases.

- Legal & Statutory Frameworks: 1) SC/ST (Prevention of Atrocities) Act, 1989 & Rules 1995 (Victim Relief & Protection). 2) MoSJE National Helpline Against Atrocities (NHAA 14566). 3) Digital Personal Data Protection (DPDP) Act 2023.
- Clinical Trauma & Vocal Stress Research: 1) DSM-5 PTSD & Acute Distress Clinical Metrics. 2) Scherer et al. (2013) Acoustic Markers of Speech Under Psychological Stress (Pitch tremor, jitter %, pause duration).
- Explainable AI & Time-Series Modeling: 1) Lundberg & Lee (2017) Unified Approach to Interpreting Model Predictions (SHAP). 2) Longitudinal trend velocity (dDDS/dt) forecasting models.
- Official Project Repository & Live Prototype: Full working source code, API backend & React frontend available at <https://github.com/balajikrishnan031/SIH-KEFI>